

Test Plan

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Test Plan

Scope

- Unit tests for attendance rules, scholarship eligibility, promotion eligibility and teacher payment.
- Integration tests for auth/session, RBAC and academic workflows.
- Security behavior tests for logout, expired sessions and private no-store/redirect behavior.
- Resource-scope tests for BranchDirector branch assignment, Teacher self-check-in limits, seeded password-hash login and attendance windows based on class session date.
- Cache management tests for public/private/no-store headers, memory-cache hit/miss evidence, tag invalidation and scoped report caching.

Test Cases

Area	Case	Expected Result
Auth	Login with mock Google token in test mode	User and session cookie returned.
Auth	Logout	Session is revoked and <code>/auth/me</code> returns 401.
Auth	Invalid token	Login returns 401.

Area	Case	Expected Result
Auth	Seeded password-hash login	Role-specific user logs in and <code>passwordHash</code> is not exposed.
Auth	Expired session	<code>/auth/me</code> returns 401.
Browser cache	Private page without session	<code>Cache-Control: no-store</code> and redirect to landing.
Browser cache	Private dashboard with valid session	Returns protected dashboard.
HTTP cache	Auth config	Returns public cache headers with <code>X-Cache-Policy: public-auth-config</code> .
HTTP cache	Session endpoint	Returns <code>Cache-Control: no-store</code> and <code>X-Cache-Policy: sensitive-no-store</code> .
Memory cache	Roles catalog read twice	First response is <code>X-Memory-Cache: MISS</code> ; second response is <code>HIT</code> .
Memory cache	Branch list after update	A write invalidates the cached list and the next read returns <code>MISS</code> .
Memory cache	Scoped branch summary	Repeated report read is cached per authenticated actor.
Python analytics cache	Health and protected endpoints	Health is short public cache; protected analytics responses are no-store.
RBAC	Student accesses consolidated reports	Returns 403.
Scope	BranchDirector creates data in assigned branch	Returns 201.
Scope	BranchDirector creates data in unassigned branch	Returns 403.
Scope	BranchDirector branch summary	Returns only assigned branches.
Scope	Teacher check-in for own profile	Returns 201.
Scope	Teacher check-in for another profile	Returns 403.
Attendance	Duplicate student/session record	Returns 409.

Area	Case	Expected Result
Attendance	Attendance window uses class session date	Historical sessions outside the window are excluded.
Absence	Director reviews justification	Review status is updated.
Scholarship	Attendance >=90% plus scores	Evaluation can be registered.
Promotion	B1 candidate plus scores	Evaluation can promote to B2.
Payment	Checked-out teacher hours	Amount equals hours times hourly rate.

Execution

```
cd 06Code  
npm test
```

Expected evidence:

```
Test Suites: 5 passed, 5 total  
Tests:      24 passed, 24 total
```